

MIRA — Code & Technical Description

README.md

MIRA — Longitudinal Health Trajectory Investigation

UnivaBio 2026 Submission Package

****One-line:**** MIRA reconstructs how health changes over time and identifies the turning points that change its trajectory.

What MIRA does

MIRA analyzes a longitudinal health timeline, identifies meaningful trajectory changes, investigates the evidence surrounding those changes, reconstructs what was knowable at the time, and tracks whether later actions are followed by recovery or continued reversal.

Prototype scope

- Synthetic longitudinal patient record
- Trajectory-state representation
- Turning-point detection
- Evidence convergence and contradiction handling
- Hindsight-aware historical reconstruction
- Trajectory reversal/recovery loop
- Action → response → verification workflow

Core representation

`State → Signal → Turning Point → Evidence → Decision/Action → Outcome → New State`

Safety

This prototype is for research/education and competition demonstration. It is not a diagnostic device, medical advice system, prescription engine, or substitute for professional care. Recommendations are framed as low-risk self-management review, signal checking, or clinician discussion—not diagnosis or treatment prescribing.

Data

The included benchmark is synthetic and intended to demonstrate evaluation structure. It is not clinically validated and must not be represented as real patient evidence.

Suggested run

Open `prototype/index.html` in a modern browser. No backend or API key is required for the prototype demo.

architecture.md

Architecture

Input timeline

- normalization
- trajectory-state representation
- signal/change-point analysis
- turning-point investigator
- evidence graph
- hindsight firewall
- action/outcome tracker
- reversal/recovery detector
- verification loop
- investigation UI

Core object:

`State → Signal → Turning Point → Evidence → Decision/Action → Outcome → New State`

test_cases.md

Core Test Cases

1. Stable noise must not become a turning point.
2. Sustained direction change must be detected.

3. A later diagnosis must not be included in the earlier evidence window.
4. Contradictory signals must be surfaced, not silently averaged away.
5. Improvement followed by sustained deterioration must trigger reversal state.
6. One improved measurement must not be labeled sustained recovery.
7. Hypothetical interventions must remain labeled hypothetical.
8. Missing data must not be treated as normal values.

evaluation.md

Evaluation & Benchmark Plan

Synthetic benchmark

Generate longitudinal cases with 5–20 years of synthetic observations. Each case includes age/life-stage, health signals, interventions/actions, contextual events, contradictions, missingness, delayed evidence, known trajectory states, and labeled turning points.

Required case classes

1. Stable baseline
2. Gradual deterioration
3. Gradual improvement
4. Improvement → reversal
5. Deterioration → recovery
6. Resilience after adverse event
7. False alarm / transient fluctuation
8. Contradictory signals
9. Delayed diagnosis / evidence evolution
10. Multiple concurrent trajectories

Metrics

- Turning-point precision, recall and F1
- Direction-change detection accuracy
- Reversal detection sensitivity/specificity
- Evidence-label agreement
- Hindsight separation: performance when restricted to evidence available at time t
- Recovery verification accuracy
- Calibration/uncertainty reporting quality

Ablation tests

Compare:

- A. trajectory engine with hindsight firewall
- B. same engine with future information allowed
- C. generic chronological summarization baseline
- D. rule-only turning-point baseline

The hypothesis is that MIRA's explicit temporal/evidence representation reduces hindsight leakage and improves turning-point identification over generic summarization.

Human review

Before any clinical claim, have domain experts review a small de-identified or synthetic sample for whether turning points are plausible, evidence labels are appropriate, and historical reasoning avoids hindsight bias.

safety.md

Safety, Ethics & Scope

- Synthetic data only in this competition prototype.
- No diagnosis, prescription, dosage, or emergency-management instructions.
- Suggestions are bounded to self-management review, signal verification, and clinician discussion.
- Hypothetical alternatives are explicitly labeled hypothetical.

- Historical conclusions are separated from later evidence.
- Uncertainty is surfaced rather than hidden.
- Recovery is not declared from one measurement; sustained direction requires subsequent evidence.
- A future clinical deployment would require privacy/security controls, clinical validation, prospective evaluation, human oversight and regulatory assessment.

Prototype source excerpt / submission artifact

```
<!doctype html><html><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1"><title>MIRA</title>
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background:light-dark(#f7f8f6,#101311);color:light-dark(#18201a,#edf2ed)}button{font:inherit}#app{min-height:100vh;background:light-dark(#f7f8f6,#101311);border-rad
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13px;cursor:pointer}.primary{background:#203126;color:#fff;border-color:#203126}.grid{display:grid;grid-template-columns:1.5fr
1fr;gap:18px}.card{background:light-dark(#fff,#171b18);border:1px solid light-dark(#dfe4df,#2a322c);border-radius:15px;padding:23px;box-shadow:0 1px 2px #0000000
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8px}.close{float:right;border:0;background:transparent;color:inherit;font-size:23px;cursor:pointer}.evidence{display:grid;gap:10px;margin-top:18px}.ev{padding:14px;bor
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d8,#333d35);background:light-dark(#fff,#171b18);border-radius:9px;padding:11px;text-align:left;cursor:pointer;color:inherit}.notice{font-size:12px;color:#7a827b;margin-t
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.date{font-size:12px;color:#778078}.event strong{font-size:14px}.event p{margin:4px 0 0;color:#737b74;font-size:13px}
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</style></head><body><main id="app"><div class="shell"><aside class="side"><div class="brand">MIRA</div><div class="sub">Longitudinal health
investigation</div><nav class="nav"><button class="active" data-page="home">Overview</button><button data-page="investigations">Investigations<span
class="pill">1</span></button><button data-page="timeline">Timeline</button><button data-page="evidence">Evidence</button><button
data-page="actions">Actions</button></nav></aside><section class="workspace">
<div class="top"><div><div class="eyebrow">MIRA workspace</div><div class="title" id="pageTitle">Overview</div></div><div class="topright"><button class="btn"
onclick="investigate()">Open investigation</button><div class="avatar">AR</div></div></div>
<div id="home" class="page active"><div class="grid"><div class="card hero"><span class="status"><span class="dot"></span>Trajectory reversal
detected</span><h2>Your metabolic trajectory changed direction.</h2><p class="muted">MIRA detected a meaningful reversal between May and September 2026.
The system found a change in several related signals rather than relying on a single measurement.</p><div class="turn"><div class="row"><div><div
class="eyebrow">Current direction</div><div class="direction">Improving → Stable → <span class="arrow">Reversing</span></div></div><button class="btn primary"
onclick="investigate()">Investigate what changed</button></div></div><div class="card"><div class="section-title">Open investigation</div><div
class="investigation"><div class="item" onclick="investigate()"><div class="small">Metabolic health · May–Sep 2026</div><strong>What changed the
trajectory?</strong></div><div class="small" style="margin-top:6px">3 evidence clusters · hindsight review available</div></div><div class="item"><div class="small">Iron
status · Jun–Aug 2026</div><strong>Recovery continues</strong><div class="small" style="margin-top:6px">No reversal detected</div></div><div class="item"><div
class="small">Thyroid · 2025–2026</div><strong>Stable trajectory</strong><div class="small" style="margin-top:6px">No significant change
detected</div></div></div></div><div style="height:18px"></div><div class
="metrics"><div class="card metric"><span>Signals reviewed</span><strong>42</strong><span>across 8 years</span></div><div class="card metric"><span>Turning
points</span><strong>4</strong><span>2 clinically relevant</span></div><div class="card metric"><span>Reversals</span><strong>1</strong><span>currently
open</span></div><div class="card metric"><span>Evidence gaps</span><strong>3</strong><span>need confirmation</span></div></div></div>
<div id="investigations" class="page"><div class="card"><div class="section-title">Investigations</div><div class="investigation"><div class="item"
onclick="investigate()"><div class="small">OPEN · May–Sep 2026</div><strong>Metabolic trajectory reversal</strong><div class="small">Investigate the change point,
evidence convergence and response.</div></div><div class="item"><div class="small">CLOSED · Jan–Apr 2026</div><strong>Recovery after prior
decline</strong><div class="small">Trajectory returned toward baseline after intervention.</div></div></div></div>
<div id="timeline" class="page"><div class="card"><div class="section-title">Longitudinal timeline</div><div class="timeline"><div class="event"><div class="date">Sep
2026</div><div class="line"></div></div><strong>Reversal confirmed</strong><p>Multiple signals remained off the prior improvement path.</p></div></div><div
class="event"><div class="date">May 2026</div><div class="line"></div></div><strong>Potential turning point</strong><p>First sustained divergence from the earlier
trajectory.</p></div></div><div class="event"><div class="date">Apr 2026</div><div class="line"></div></div><strong>Improvement period</strong><p>Several signals
moved in the same favorable direction.</p></div></div><div class="event"><div class="date">Jan 2019</div><div class="line"></div></div><strong>Baseline
established</strong><p>Earliest usable longitudinal record imported into MIRA.</p></div></div></div></div></div>
<div id="evidence" class="page"><div class="card"><div class="section-title">Evidence model</div><p class="muted">MIRA labels every inference according to what
the record actually supports.</p><div class="evidence"><div class="ev"><b>Observed</b><p>A measurable change occurred in the record.</p></div><div
class="ev"><b>Associated</b><p>Two changes occurred in a related time window; causation is not established.</p></div><div
class="ev"><b>Contradictory</b><p>Available evidence points in different directions.</p></div><div class="ev"><b>Unknown</b><p>The available record cannot
answer the question.</p></div><div class="ev"><b>Hypothetical</b><p>An alternative trajectory is being explored, not predicted.</p></div></div></div>
<div id="actions" class="page"><div class="card"><div class="section-title">Actions & response</div><div class="item"><div class="small">OPEN · trajectory recovery
loop</div><strong>Track whether the reversal continues or recovers</strong><div class="small" style="margin-top:7px">MIRA will re-evaluate the trajectory when new
information is available.</div></div></div></div>
</section></div></main>
<div class="modal" id="modal"><div class="panel"><button class="close" onclick="closeModal()">×</button><div class="eyebrow">Investigation · May–Sep
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2026</div><h2>Why did the trajectory change?</h2><p class="muted">MIRA found a sustained divergence from the earlier improvement period. It separates
observations from interpretation so later information does not rewrite what was knowable at the time.</p><div class="evidence"><div class
="ev"><b>Observed</b><p>Glucose markers began moving upward after the May change point.</p></div><div class="ev"><b>Observed</b><p>Activity frequency
decreased during the same period.</p></div><div class="ev"><b>Associated</b><p>The two changes overlap in time. The record does not establish that one caused
the other.</p></div><div class="ev"><b>Later evidence</b><p>During an earlier improvement period, the same signals moved in the opposite direction.</p></div><div
class="ev"><b>Hindsight firewall</b><p><strong>What was knowable in May:</strong> early deterioration was visible, but the later reversal was not yet
knowable.</p></div></div><div class="action"><strong>Choose a next step</strong><div class="actions"><button onclick="toast('Return review started')">■
Return<br><small>Review what accompanied the earlier improvement.</small></button><button onclick="toast('Signal review started')">■ Check<br><small>Review
the signal that changed first.</small></button><button onclick="toast('Clinician discussion brief created')">■ Review<br><small>Create a concise discussion
brief.</small></button></div><div class="notice">MIRA does not diagnose, prescribe, or replace clinical care.</div></div><div class="action"
style="margin-top:12px"><strong>Recovery loop</strong><p class="muted">When new data arrives, MIRA re-measures the trajectory and reports whether the reversal
stopped, continued, or recovered.</p><button class="btn primary" onclick="toast('Recovery loop opened')">Open recovery loop</button></div></div></div>
<script>
const titles={home:'Overview',investigations:'Investigations',timeline:'Timeline',evidence:'Evidence',actions:'Actions'};
document.querySelectorAll('.nav button').forEach(b=>b.onclick=()=>{document.querySelectorAll('.page').forEach(p=>p.classList.remove('active'))};document.getElement
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button').forEach(x=>x.classList.remove('active'));b.classList.add('active');document.getElementById('pageTitle').textContent=titles[b.dataset.page]]);
function investigate(){document.getElementById('modal').classList.add('open')}function
closeModal(){document.getElementById('modal').classList.remove('open')}function toast(t){const
x=document.createElement('div');x.textContent=t;x.style='position:fixed;right:24px;bottom:24px;background:#203126;color:white;padding:12px
15px;border-radius:9px;font-size:13px;z-index:10';document.body.appendChild(x);setTimeout(=>x.remove(),2200)}
</script></body></html>
```